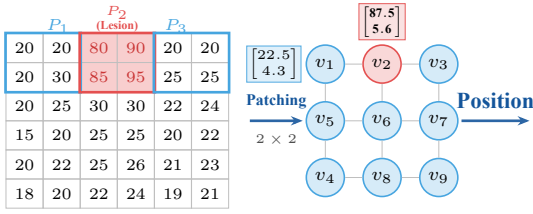


Stage 1: Raw Image Matrix → Patch Embedding & Positional Encoding

Raw Image $I \in \mathbb{R}^{6 \times 6}$



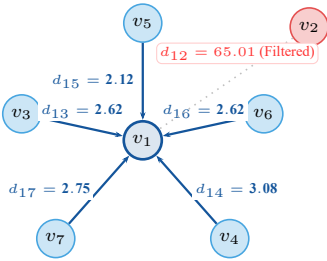
Stem Embedding & Position Encoding:

$$\hat{\mathbf{x}}_i = \text{Stem}(\mathbf{x}_i) + \mathbf{p}_i, \quad \mathbf{p}_i \in \mathbb{R}^2$$

$$\begin{aligned} \hat{\mathbf{x}}_1 &= [22.5, 4.3]^\top, & \hat{\mathbf{x}}_2 &= [87.5, 5.6]^\top \text{ (Lesion)} \\ \hat{\mathbf{x}}_3 &= [20.0, 3.5]^\top, & \hat{\mathbf{x}}_4 &= [25.0, 2.5]^\top \\ \hat{\mathbf{x}}_5 &= [21.0, 5.8]^\top, & \hat{\mathbf{x}}_6 &= [24.5, 6.0]^\top \\ \hat{\mathbf{x}}_7 &= [21.0, 2.0]^\top, & \hat{\mathbf{x}}_8 &= [20.0, 1.4]^\top \\ \hat{\mathbf{x}}_9 &= [22.0, 1.8]^\top \end{aligned}$$

(S1: Dynamic Graph Construction)

Stage 2: Dynamic K -NN Metric Space Graph Construction ($K = 5$)

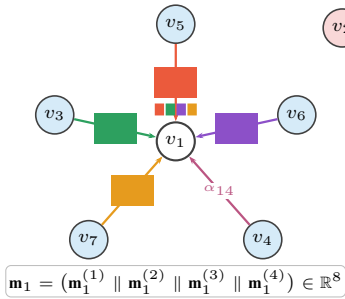


Metric Space Distance Ranking at v_1 ($K = 5$):

1. $d_{15} = \|\hat{\mathbf{x}}_5 - \hat{\mathbf{x}}_1\|_2 = 2.12$ (Rank 1 - Closest)
 2. $d_{13} = \|\hat{\mathbf{x}}_3 - \hat{\mathbf{x}}_1\|_2 = 2.62$, 3. $d_{16} = \|\hat{\mathbf{x}}_6 - \hat{\mathbf{x}}_1\|_2 = 2.62$
 4. $d_{17} = \|\hat{\mathbf{x}}_7 - \hat{\mathbf{x}}_1\|_2 = 2.75$, 5. $d_{14} = \|\hat{\mathbf{x}}_4 - \hat{\mathbf{x}}_1\|_2 = 3.08$
- Filtered Outliers: $d_{12} = 65.01$ (Lesion Outlier), $d_{18} = 3.83$
 $\Rightarrow$ Dynamic Neighborhood: $\mathcal{N}_5(v_1) = \{v_5, v_3, v_6, v_7, v_4\}$

(S2: Multi-Head GATConv Attention)

Stage 3: Multi-Head Dynamic GATConv Attention Aggregation ($K = 5$, $H = 4$ Heads)



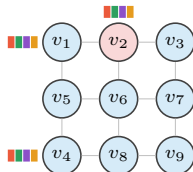
Multi-Head Attention Aggregation ($H = 4$, $K = 5$):

$$\begin{aligned} \text{Head 1 } (\alpha^{(1)}): \mathbf{m}_1^{(1)} &= \sum_j \alpha_{1j}^{(1)} \mathbf{d}_{1j} = [-0.650, +0.065]^\top \\ \text{Head 2 } (\alpha^{(2)}): \mathbf{m}_1^{(2)} &= \sum_j \alpha_{1j}^{(2)} \mathbf{d}_{1j} = [-0.575, -0.155]^\top \\ \text{Head 3 } (\alpha^{(3)}): \mathbf{m}_1^{(3)} &= \sum_j \alpha_{1j}^{(3)} \mathbf{d}_{1j} = [-0.250, +0.010]^\top \\ \text{Head 4 } (\alpha^{(4)}): \mathbf{m}_1^{(4)} &= \sum_j \alpha_{1j}^{(4)} \mathbf{d}_{1j} = [-0.325, -0.425]^\top \end{aligned}$$

$$\Rightarrow \mathbf{m}_1 = (\mathbf{m}_1^{(1)} \parallel \mathbf{m}_1^{(2)} \parallel \mathbf{m}_1^{(3)} \parallel \mathbf{m}_1^{(4)}) = [-0.650, +0.065, \dots, -0.425]^\top \in \mathbb{R}^8$$

(S3: Grapher-FFN Block Update)

Stage 4: State Update & Residual FFN ($\times L = 8$ Blocks)



Residual State Update ($\times L = 8$ Blocks) at v_1 :

$$\mathbf{h}_1^{(1)} = \text{GELU}\left(\text{BN}(\mathbf{W}_{\text{out}}[\mathbf{x}_1^{(0)} \parallel \mathbf{m}_1] + \mathbf{b}_{\text{out}})\right) = [+0.42, -0.18]^\top$$

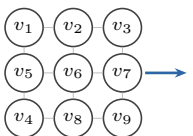
$$\tilde{\mathbf{x}}_1^{(1)} = \mathbf{x}_1^{(0)} + \mathbf{h}_1^{(1)} = [22.5, 4.3]^\top + [0.42, -0.18]^\top = [22.92, 4.12]^\top$$

$$\mathbf{x}_1^{(1)} = \tilde{\mathbf{x}}_1^{(1)} + \text{FFN}(\tilde{\mathbf{x}}_1^{(1)}) = [23.10, 4.05]^\top$$

$$\Rightarrow \text{Iterated through } L = 8 \text{ blocks} \rightarrow \mathbf{x}_1^{(8)} = [18.50, 2.40]^\top$$

(S4: Global Readout & Prediction)

Stage 5: Global Readout (GAP) → Final Output Classification



Global Readout & Softmax:

$$\mathbf{z} = \frac{1}{9} \sum_{v=1}^9 \mathbf{x}_v^{(8)} = [16.92, 2.17]^\top$$

$$\mathbf{l} = \mathbf{W}_{\text{cls}} \mathbf{z} + \mathbf{b}_{\text{cls}} = [2.788, 0.000]^\top$$

$$\hat{\mathbf{y}} = \text{Softmax}(\mathbf{l}) = [94.2\%, 5.8\%]^\top$$

Final Prediction ($\hat{\mathbf{y}}$):

1. Healthy Leaf: 94.2% █
2. Late Blight: 5.8% █

Diagnostic Decision:

HEALTHY LEAF (94.2%)